CTGGAN: Reliable Fetal Heart Rate Signal Generation Using GANs

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July 1, 2024



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Background

- **Prenatal care** is when you get checkups from a doctor, nurse, or midwife throughout your pregnancy. It helps keep you and your future baby healthy.
- **Fetal monitoring** is a routine procedure which enables the clinician to monitor the wellbeing of the fetus and to intervene if appropriate.

- **Question: Do you know what fetal monitoring involve?**

Background

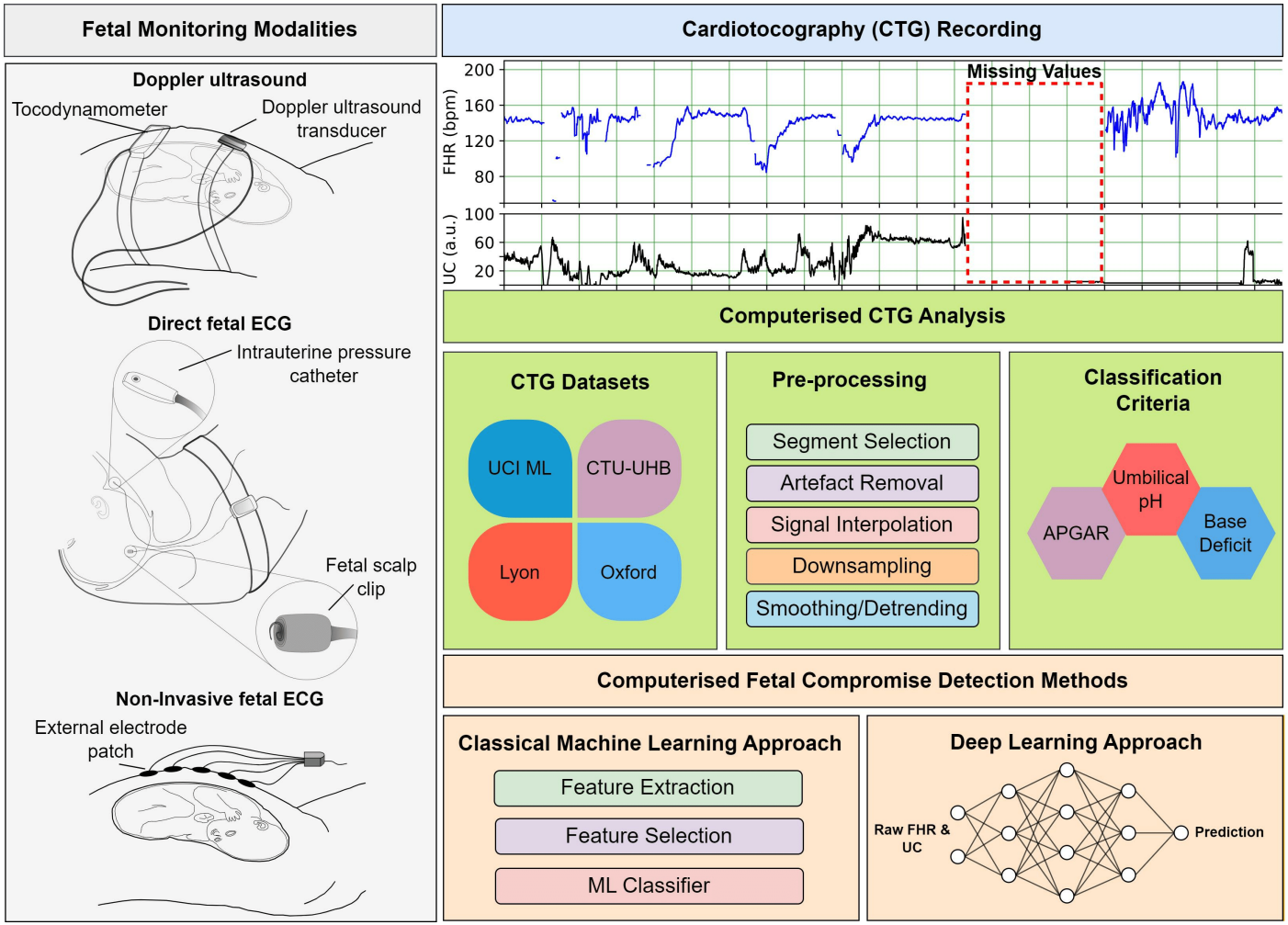


Figure: Computerised Analysis of Cardiotocography (CTG). Thanks to: Mendis, Lochana, et al. "Computerised cardiotocography analysis for the automated detection of fetal compromise during labour: a review." Bioengineering 10.9 (2023): 1007.

Background

- Fetal Monitoring: Cardiotocography (CTG)
  - Analyzing fetal heart rate (FHR) and maternal uterine contractions
  - Assisting healthcare professionals in maternal and fetal care
- FHR monitoring insights
  - Fetus's physiological state and stress responses
  - Early detection of anomalies
- Subjective interpretation challenges
  - Inter-observer and intra-observer variability

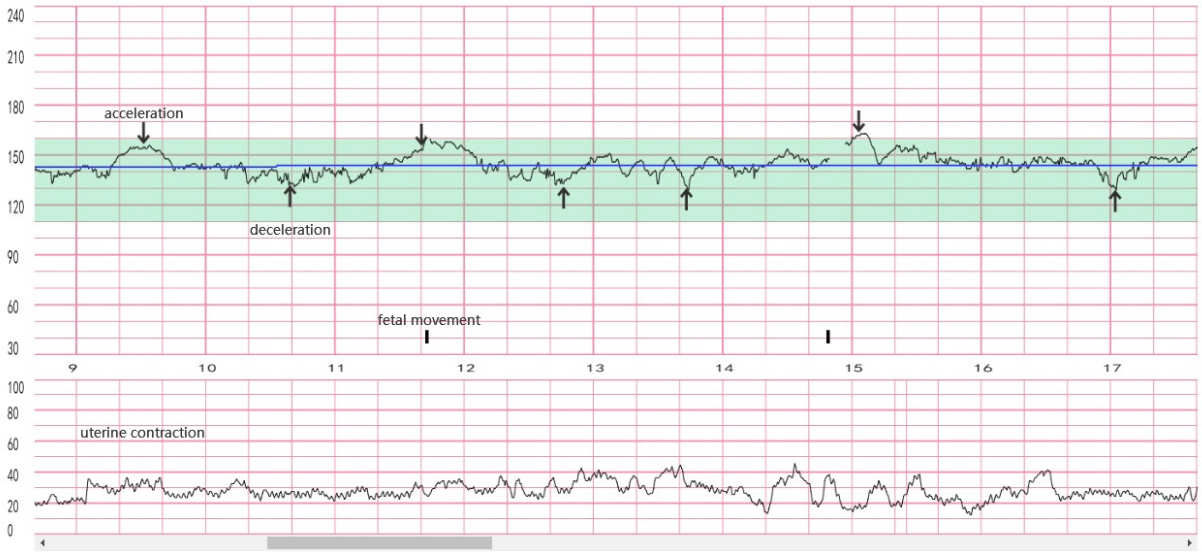


Figure: A CTG signal.

Challenges

- Deep learning potential for automated FHR interpretation
  - Scarcity of labeled FHR datasets
- CTU-UHB database
  - Only 552 recordings
  - Limited data availability
- Generative models to address data scarcity
  - Generative adversarial networks (GANs)
  - Generating synthetic FHR signals
- Benefits of synthetic data generation
  - Nearly unlimited training data
  - Ensuring patient confidentiality

Contributions

- Introduction of CTGGAN
  - Novel Generative Adversarial Network for synthetic FHR signals
- Conditional GANs (CGANs) with label training
  - Generation of samples in various categories
- 1D transposed convolutional layers
  - Capture intricate patterns and high variability in FHR signals
- State-of-the-art techniques
  - Self-attention mechanisms: complex and long-range dependency
  - Residual block: improve training efficiency
- Advanced loss function
  - Combination of Wasserstein distance with gradient penalty
  - Improved training stability and data diversity
- Superior fidelity and diversity
  - Excels existing SOTA methods
- Addressing data scarcity and class imbalance
  - Advanced data-driven approaches in prenatal care

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Oversampling Methods

1. Random flipping, translation, cropping, and affine transformations
2. Synthetic Minority Over-sampling Technique (SMOTE) algorithm

References


Branco, Paula et al, A Survey of Predictive Modeling on Imbalanced Domains, *ACM Computing Surveys*, 49(2):1–50, 2016.


Chawla, Nitesh V. et al., SMOTE: Synthetic Minority Over-sampling Technique,*Journal of Artificial Intelligence Research*, 16:321–357, 2022.

Mathematical Methods

1. Nonlinear model
2. Artificial vector model

References


Clifford, GD and McSharry, PE, Generating 24-hour ECG, BP and respiratory signals with realistic linear and nonlinear clinical characteristics using a nonlinear model, *Computers in Cardiology*, 2004.


Clifford, GD et al., An artificial vector model for generating abnormal electrocardiographic rhythms, *Physiological Measurement*, 31(5):595, 2010.

Deep Learning Methods

1. Variational Autoencoders (VAEs)
2. Generative Adversarial Networks (GANs)
3. Diffusion Models

References


Pastor-Serrano, Oscar et al., A semi-supervised autoencoder framework for joint generation and classification of breathing,*Computer Methods and Programs in Biomedicine*, 209:106312, 2021.


Zhang, Yefei et al, FHRGAN: Generative adversarial networks for synthetic fetal heart rate signal generation in low-resource settings, *Information Sciences*, 594:136–150, 2022.


Ahmed, Halal Abdulrahman, et al., Generating Synthetic Fetal Cardiotocography Data with Conditional Generative Adversarial Networks. SOCO 2023, pp.111-120.

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Architecture of CTGGAN

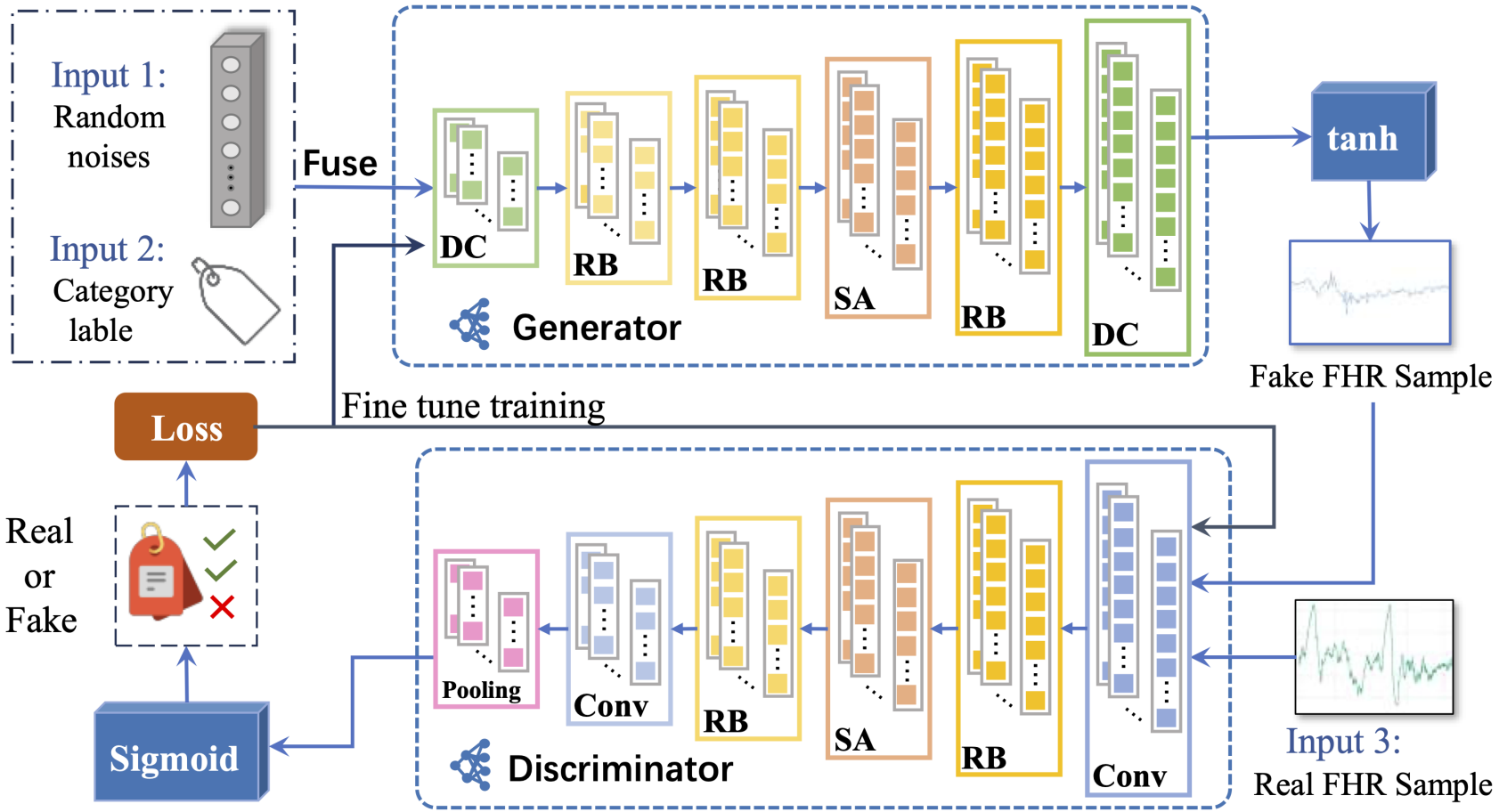


Figure: CTGGAN for FHR signal generation.

Self-Attention Block in CTGGAN

- Self-Attention mechanism enhances focus on specific regions of the input feature map
- Enables effective capture and processing of long-range dependencies
- Crucial for understanding complex data patterns

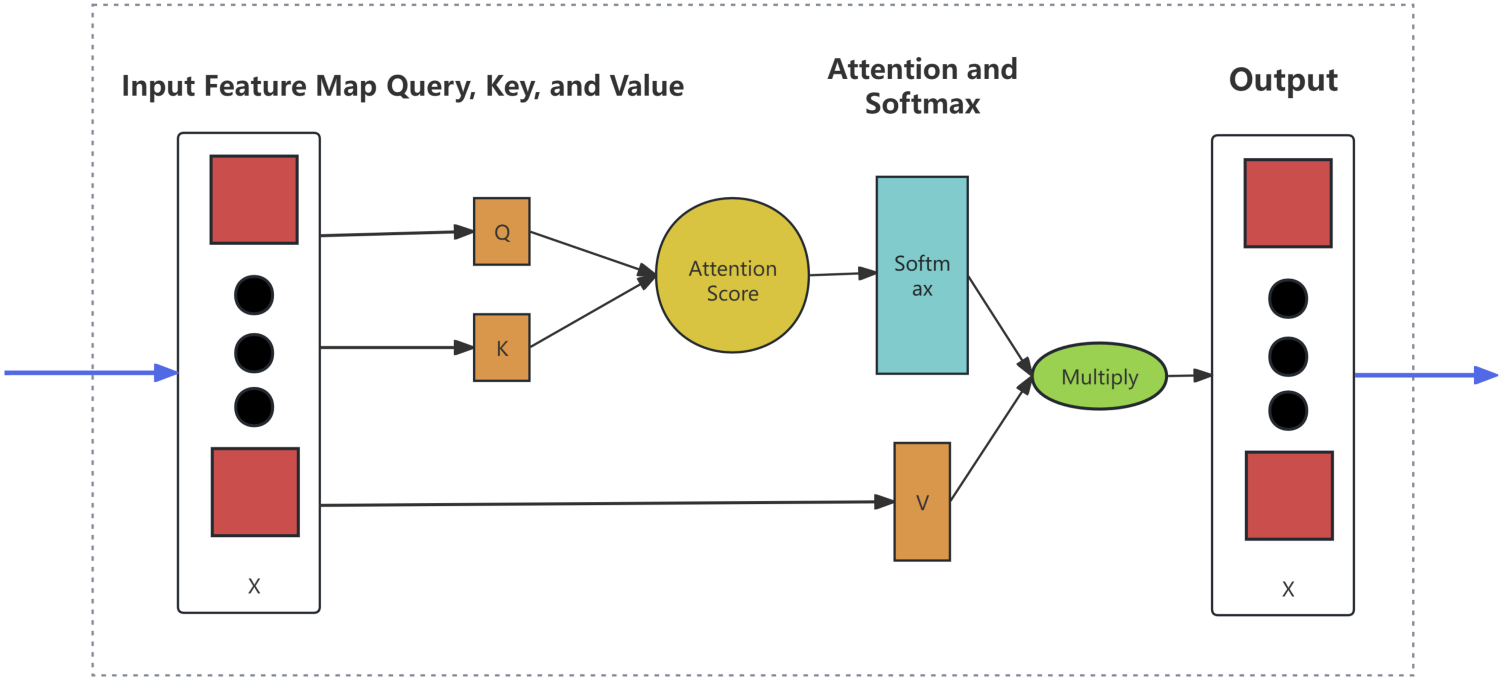


Figure: Self-Attention Block in CTGGAN.

Wasserstein Distance

$$W(P_r, P_g) = \inf_{\gamma \in \prod(P_r, P_g)} \mathbb{E}_{(x,y)\sim\gamma}[\|x - y\|] \tag{1}$$

- P_r is the real data distribution.
- P_g is the generated data distribution.
- $\prod(P_r, P_g)$ denotes the set of all joint distributions $\gamma(x, y)$ that have marginals P_r and P_g.
- x is a sample from P_r.
- y is a sample from P_g.
- $\|x - y\|$ represents the distance between samples x and y.



Wasserstein GAN with Gradient Penalty Term

In training, the discriminator D minimizes the Wasserstein distance, fixed by θ_G^*:

$$L_D(\theta_D, \theta_G^*) = W(P_r, P_f) \tag{2}$$

Similarly, the generator G is trained to minimize the Wasserstein distance:

$$L_G(\theta_G, \theta^* D) = \mathbb{E}_{x_f \sim P_f}[D_{\theta^* D}(x_f)] = \mathbb{E}_{x_f \sim P_f}[D_{\theta_D}(G_{\theta_G^*}(z))] \tag{3}$$

WGAN-GP resolves this by imposing Lipschitz continuity on the discriminator more effectively. It includes a gradient penalty in the loss function, as follows:

$$W(P_r, P_f) = \overline{W}(P_r, P_f) + \lambda \mathbb{E}_{\hat{x} \sim \hat{P}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \tag{4}$$

$\hat{x}$ represents samples along linear interpolations between P_r and P_f, and λ is a hyperparameter that balances the original WGAN loss and the gradient penalty.



Loss Function

$$L_G = -\mathbb{E}_{\tilde{x}\sim P_g}[D(\tilde{x})] + \alpha \cdot \|F_r - F_g\|_2^2 \tag{5}$$

- L_G represents the loss function of the generator.
- $\tilde{x}$ denotes the samples generated by the generator, following the distribution P_g.
- $D(\tilde{x})$ is the discriminator's output for the generated samples, indicating the probability of the sample being real.
- The term $\alpha \cdot \|F_r - F_g\|_2^2$ is the feature matching component, where:
  - α is a weighting factor that balances the importance of the feature matching component in the overall loss function.
  - $\|F_r - F_g\|_2^2$ is the squared L2 norm, which measures the distance between feature representations of real samples (F_r) and generated samples (F_g) extracted from an intermediate layer of the discriminator.

Configuration of CTGGAN

Table: Architectural Details of the Generator and Discriminator for FHR Signal Data Augmentation. Note: LReLU is LeakyReLU with an alpha value of 0.2; BN: Batch Normalization; Conv denotes convolution; SA: Self-Attention Block; RB: Residual Block; FC: Fully Connected Layer.

Generator			Discriminator		
Layer	Structure	Output Shape	Layer	Structure	Output Shape
Input	-	(3, 100)	Input	-	(3, 1000)
DC1	Conv1d-ReLu-BN	(256, 125)	Conv1	Conv1d-LReLu	(32, 1000)
RB2	(Conv1d-ReLu-BN)*2	(128, 250)	RB2	Conv1d-LReLu-BN	(64, 500)
SA3	Self-Attention	(128, 250)	SA3	Self-Attention	(64, 500)
RB4	Conv1d-ReLu-BN	(64, 500)	RB4	Conv1d-LReLu-BN	(128, 250)
RB5	Conv1d-ReLu-BN	(32, 1000)	RB5	Conv1d-LReLu	(256, 125)
DC6	Conv1d-Tanh	(1, 1000)	FC6	Pooling-Sigmoid	(1, 1)



Training Procedure

Table: Training Procedure of CTGGAN

CTGGAN Training Procedure
Input: Real FHR data samples from CTG dataset $x \sim P_{\text{data}}$ and latent vectors z of dimension 100 **Output:** Fully trained generator model G_h
1. Initialize gradient penalty coefficient $\lambda = 10$, epochs $n_{\text{epochs}} = 150$, batch size $m = 32$, critic-to-generator iteration ratio $n_{\text{critic}} = 5$, Adam parameters: $\alpha = 0.0001$, $\beta_1 = 0$, $\beta_2 = 0.9$. Initialize generator G and discriminator D parameters using a truncated normal distribution, variance 0.02
for $j \leftarrow 1$ **to** n_{epochs} **do**   **for** $t \leftarrow 1$ **to** n_{critic} **do**   **for** each m-sized batch **do**   Sample real FHR data $x \sim P_{\text{data}}$ and noise vectors $z \sim \mathcal{N}(0, 1)$ $\hat{x} \leftarrow G(z)$ Calculate discriminator loss L_D from real and generated data   Update discriminator D via Adam optimizer with α, β_1, β_2 **end for** $\omega \leftarrow \text{Adam}\left(\nabla \frac{1}{m} \sum_{i=1}^{m} L_D, \omega, \alpha, \beta_1, \beta_2, \varepsilon\right)$ **end for**   Sample a batch of noise matrices $z^{(i)}{}_{i=1}^{m} \sim \mathcal{N}(0, 1)$ Determine generator loss L_G Update generator G using Adam optimizer and parameters α, β_1, β_2 **end for**
Output: Store the trained CTGGAN model G_h

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Dataset

- CTU-UHB dataset from Czech Technical University (CTU) in Prague and University Hospital in Brno (UHB)
- Contains 552 CTG recordings
  - Selected from 9,164 recordings (2010-2012)
- CTG recordings include:
  - FHR (Fetal Heart Rate)
  - Uterine contraction signals
  - Sampled at 4Hz over 90 minutes
- Neonatal umbilical artery pH as gold standard
  - pH $\geq$ 7.15: Normal, 447 normal FHRs
  - pH $\leq$ 7.15: Pathological, 105 pathological FHRs
- Study sample details:
  - 200 real FHR samples
  - 100 normal, 100 pathological
  - 8:2 ratio for training and testing sets

Pre-processing

- Discarding data where FHR was zero for $>$ 15s, or FHR was $<$ 50 bpm or $>$ 200 bpm.
- Replacing abnormalities with Hermite interpolation.

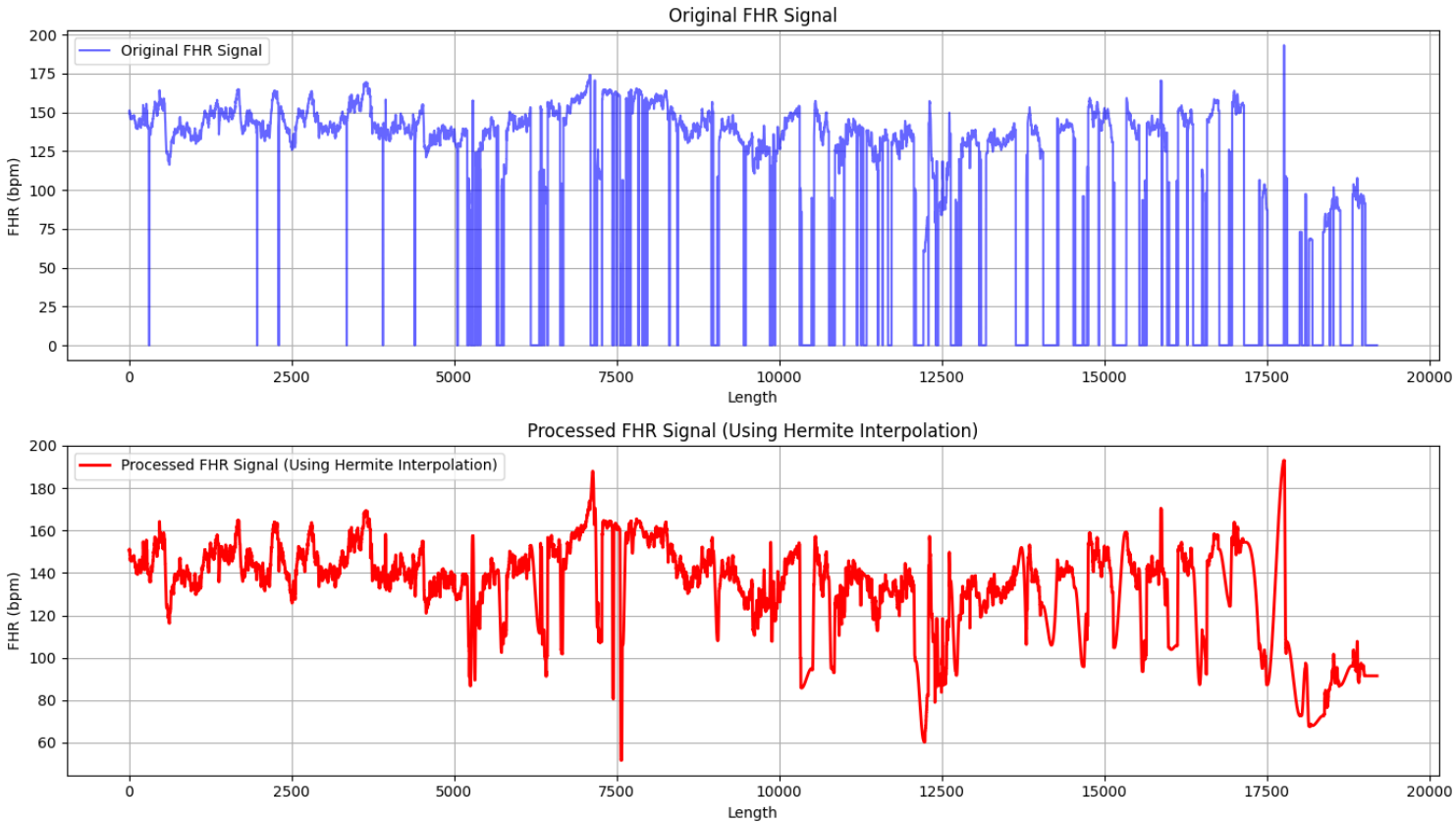
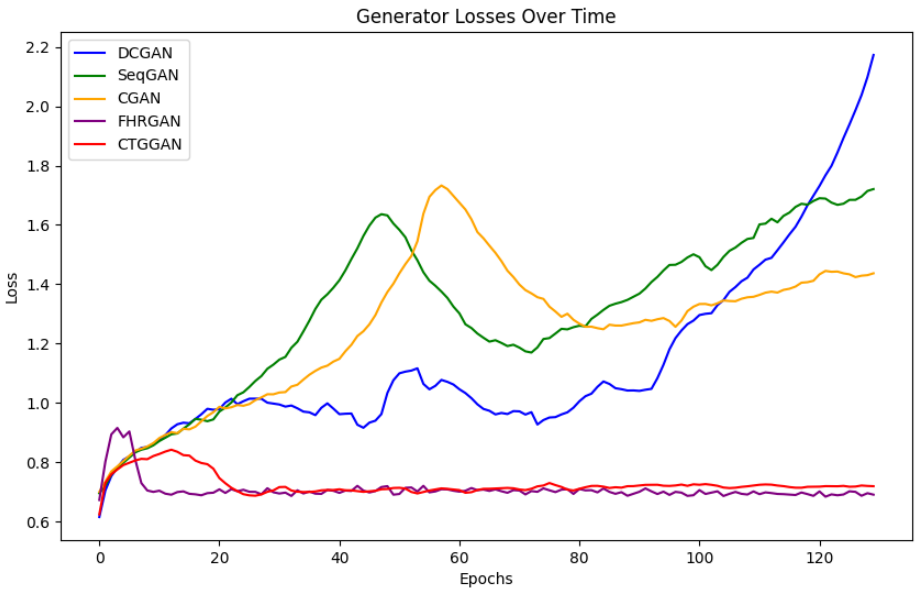
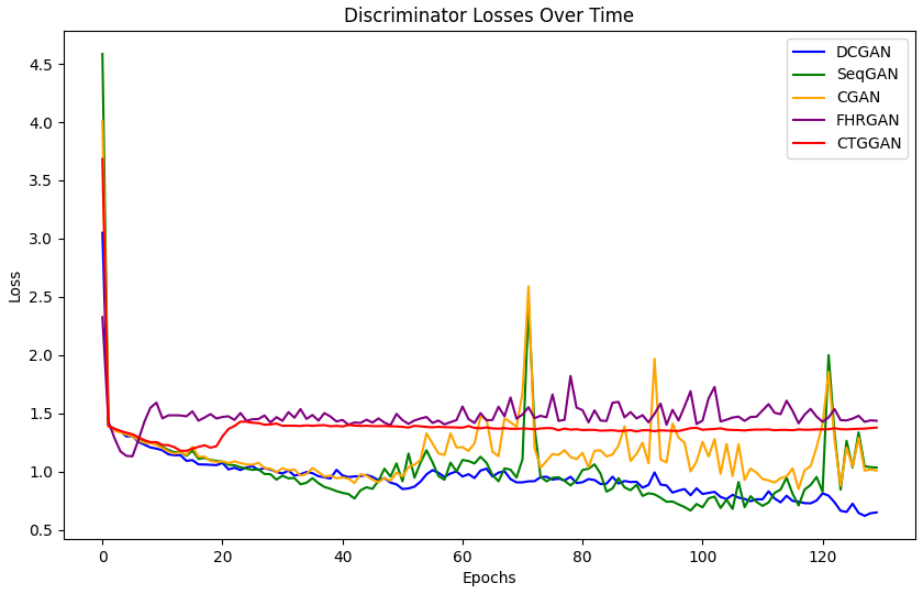


Figure: FHR Signals Before and After Data Pre-processing.

Training Performance



(a) Comparative Generator Loss across Different Models



(b) Comparative Discriminator Loss across Different Models

Figure: Comparative Losses of Generator and Discriminator across Different Models

Experimental Results

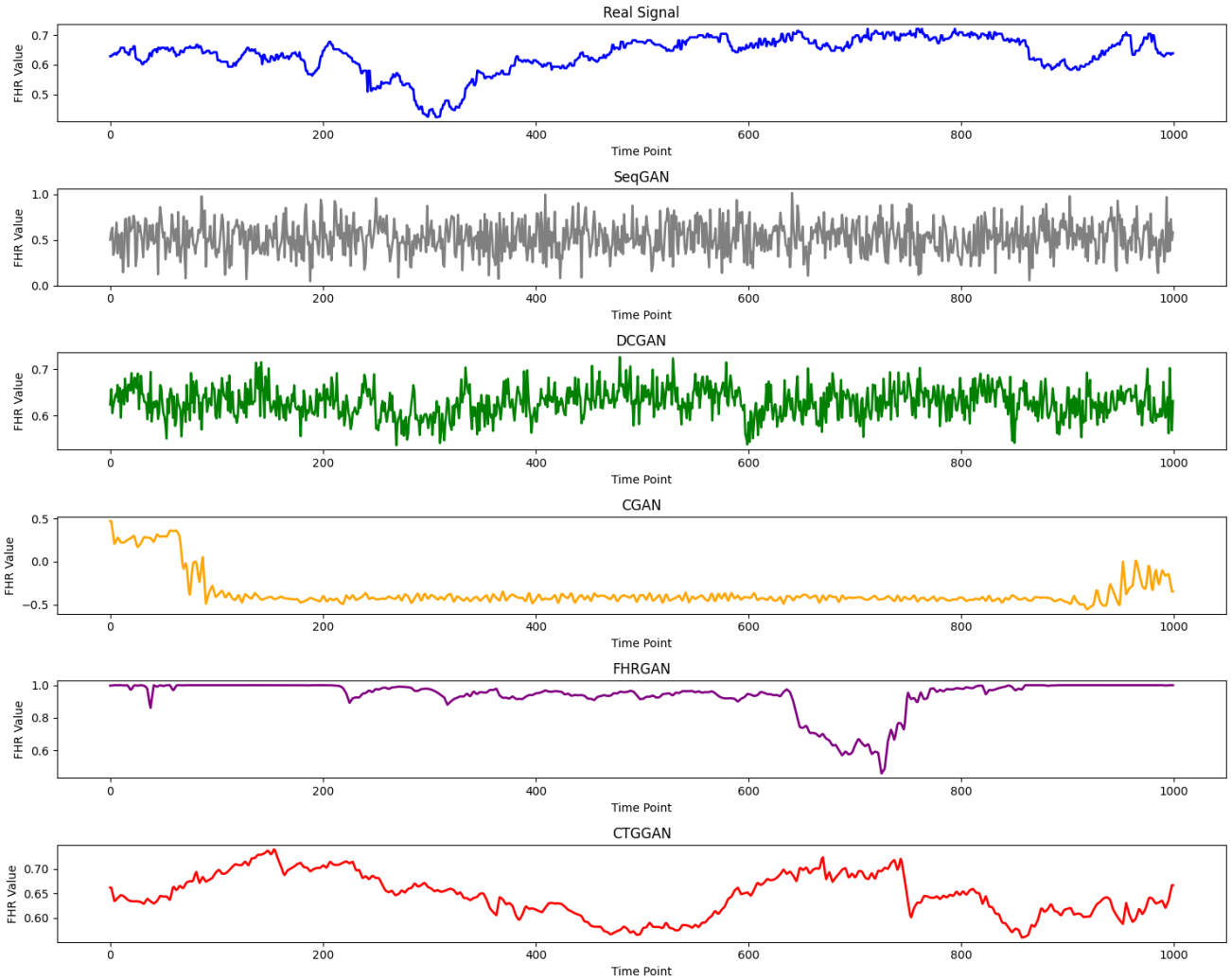


Figure: Visual Comparison of FHR Signals Generated by GANs.

Evaluation Metrics

- **Reconstruction Error (RE):** measure the difference between the original input data and the data reconstructed by a model
- **Field Distance (FD):** measure the similarity between two distributions by finding the minimum possible distance between the two continuous signals
- **Maximum Mean Discrepancy (MMD):** measure the difference between the distributions based on their means in a reproducing kernel Hilbert space (RKHS)
- **Sliced Wasserstein Distance (SWD):** averaging the Wasserstein distance over multiple 1-dimensional projections
- **Percent Root Mean Square Difference (PRD):** the relative difference between two sets of signals



Experimental Results

Table: Comparison of Evaluation Metrics for Different Models

Model	**Evaluation Metrics**				
	RE (↓)	FD (↓)	MMD (↓)	SWD (↓)	PRD (↓)
SeqGAN	10.324	0.974	6.527	0.541	10.635
DCGAN	9.532	0.795	5.267	0.254	10.431
CGAN	9.012	0.763	5.022	0.254	9.431
FHRGAN	8.059	0.682	1.975	0.086	5.437
CTGGAN	**5.621**	**0.614**	**0.215**	**0.012**	**4.821**

Discussion

- Presence of certain limitations
- Effective overall variability capture
- Shortfall in emulating specific FHR signal characteristics, such as sinusoidal patterns
- Potential for enhancing generalization capabilities
  - Utilization of larger, real-world dataset
  - Other advanced generative models: diffusion models
- Promising application of GANs for FHR signal generation

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Conclusions and Future Work

- Introduction of CTGGAN for synthetic FHR signal generation
- Integration of:
  - Customized Self-Attention mechanism
  - Residual network architecture
  - Optimized loss function
- Achieves SOTA performance in producing high-fidelity FHR traces
- Excels existing GAN models in:
  - Mimicking real FHR signal characteristics
  - Offering superior sample quality and diversity
  - Maintaining stable training
- Significant contribution to:
  - Overcoming data scarcity
  - Addressing class imbalance in FHR research and clinical practice

Contacts

- Source Code: https://github.com/ijcnn2024/CTGGAN
- Research Group Website: https://lvyupku.github.io/
- Email: lvyu@sztu.edu.cn

